

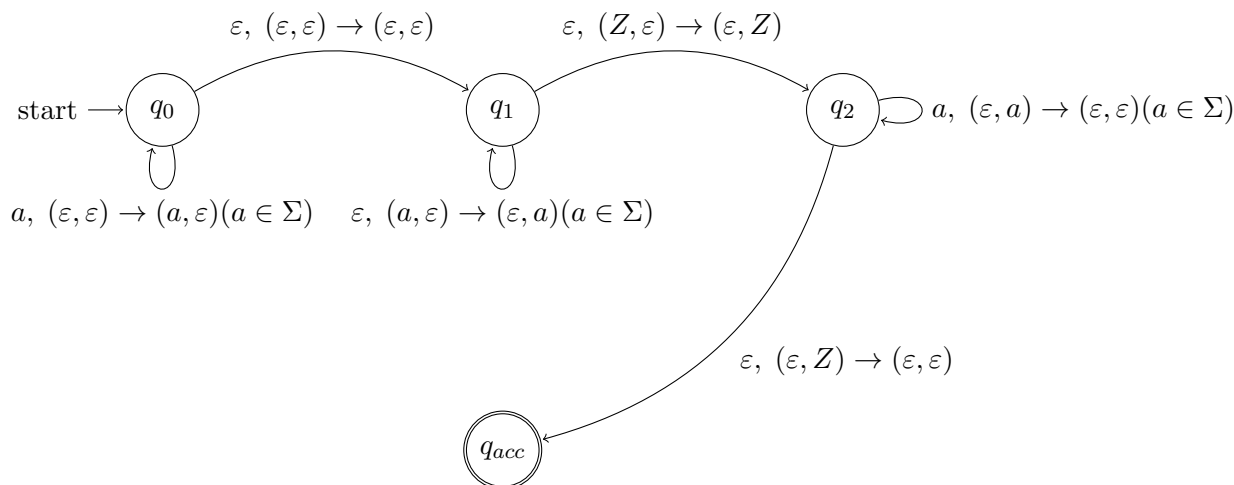
Homework 7G: Church-Turing Thesis and Turing-Acceptability

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Due: 3/10/2025

You should submit a typeset or *neatly* written pdf on Gradescope. The grading TA should not have to struggle to read what you've written; if your handwriting is hard to decipher, you will be asked to typeset your future assignments. Four bonus points if you use $\text{\LaTeX}$, and our template. You may collaborate with other students, but any written work should be your own. **Many of these problems have two sentence answers if you can apply the theorems proved in class.**

1. Suppose we have a push down automata with two stacks. It has nondeterministic transitions of the form **read**, (**pop1**, **pop2**) $\rightarrow$ (**push1**, **push2**). It reads from the input, pops from both stacks, and pushes to both stacks simultaneously. Give the state diagram of this PDA with two stacks to decide $\{ww \mid w \in \Sigma^*\}$. (Hint: Think about what allowed a single stack PDA to decide ww^R but not ww ?)



2. Prove that there exists an undecidable subset of 1^* .

In lecture, we established there are countably many Turing Machines. Since there are only countably many Turing Machines, there are only countably many decidable languages. However, since 1^* is a one-to-one correspondence with $\mathbf{N}$, the set of all its subsets has cardinality $2^{\mathbf{N}_0}$, which is uncountable (represented in lecture as 2^S). Therefore, we find there must be some subsets of 1^* that are not decidable.

3. A **useless** state in a Turing machine is one that is never entered on any input string. Consider the problem of determining whether a Turing machine has any useless states. Show that the language of this problem is undecidable. Is it Turing Acceptable?

We will reduce from the empty language problem. Given a Turing Machine $\mathbf{M}$, we wish to construct a Turing Machine $\mathbf{M}'$ that simulates $\mathbf{M}$ for any input and enters a

state q iff $\mathbf{M}$ accepts; then q is useless in $\mathbf{M}'$ when $L(M) = \emptyset$. Additionally, we find the language of machines having a useless state is not Turing acceptable since verifying a state has never been reached requires checking an infinite number of computations. Interestingly, however, we find the complement (machines in which every state is reached on some input) is Turing acceptable since there is a finite number of states.

4. Define the language

$$L_0 = \{\langle M \rangle \mid M \text{ halts on the empty string}\}$$

Prove that L_0 is undecidable by constructing a many-to-one reduction from L_{halt} . That is, construct a computable function f such that

$$\langle M, w \rangle \in L_{\text{halt}} \iff f(\langle M, w \rangle) \in L_0.$$

We define the function f on inputs $\langle M, w \rangle$ by defining $f(\langle M, w \rangle) = \langle M' \rangle$, where M' is the Turing Machine that on the empty string ε where, on input, writes w onto its tape and simulates M on w . Since M' halts on ε iff M halts on w , we find this many-to-one reduction demonstrates $\langle M, w \rangle \in L_{\text{HALT}}$ iff $f(\langle M, w \rangle) \in L_0$, which implies that L_0 is undecidable.